

TASK 4

Sentiment Analysis

Classifying emotions in book reviews using NLP & VADER

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This report documents an NLP-based sentiment analysis pipeline applied to 300 Amazon-style book reviews. The VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon assigns compound polarity scores in the range $[-1, +1]$ to each review, classifying it as Positive, Neutral, or Negative. Results are visualised through distribution charts, word clouds, and a summary dashboard.

Methodology

1. Dataset — 300 book reviews (100 per sentiment class) written to represent realistic Positive, Neutral, and Negative opinions.

2. VADER Analysis — Each review is passed through NLTK's SentimentIntensityAnalyzer which returns four scores:

Score	Range	Meaning
pos	[0, 1]	Proportion of positive sentiment
neu	[0, 1]	Proportion of neutral sentiment
neg	[0, 1]	Proportion of negative sentiment
compound	[-1, +1]	Normalised overall polarity score

3. Classification Rule

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if compound >= +0.05: label = "Positive"
elif compound <= -0.05: label = "Negative"
else: label = "Neutral"
```

4. Text Pre-processing — Reviews are lowercased, punctuation stripped, tokenised, and stop-words removed to extract meaningful word frequencies per sentiment class.

Results Summary

Reviews Analysed 300	Positive 174	Negative 108	Neutral 18
VADER Accuracy 67.0%	Mean Compound Score +0.165	Avg Positive Score +0.585	Avg Negative Score -0.483

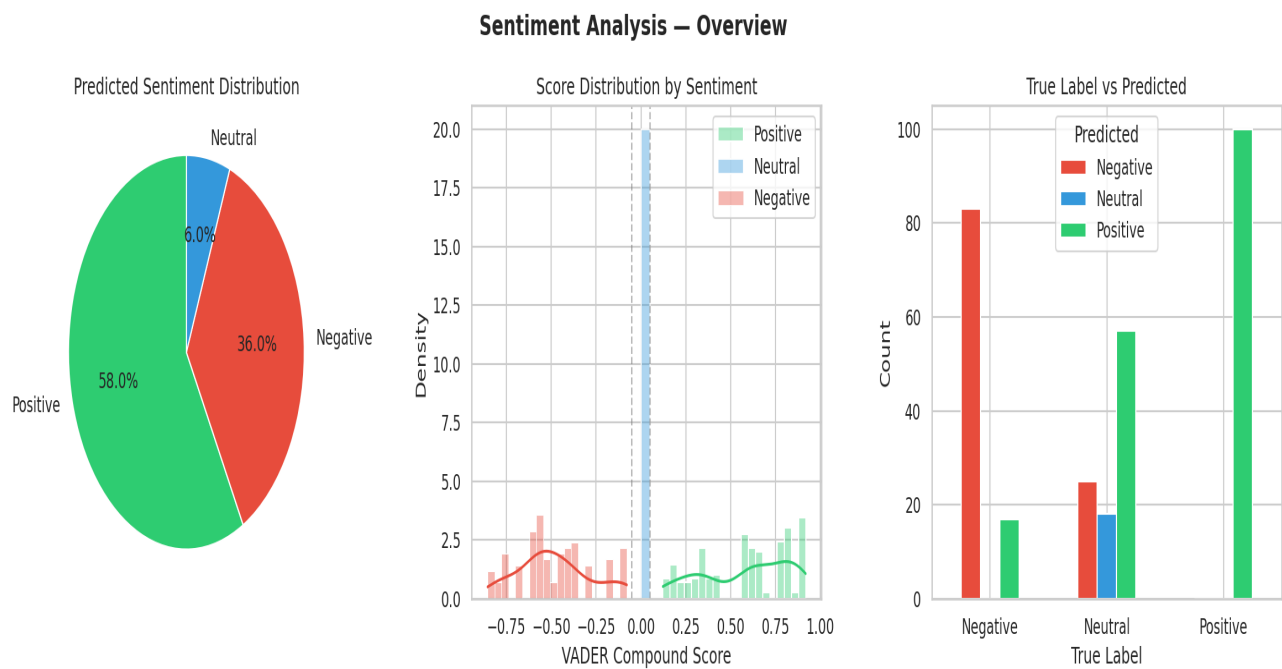
Sample Predictions

True	Predicted	Compound	Review (truncated)
Positive	Positive	+0.886	I adored this novel. The plot twists were surprising ye...
Neutral	Positive	+0.565	It was okay. Some interesting moments but nothing that ...
Positive	Positive	+0.624	Couldn't put it down. Stayed up all night to finish it...
Neutral	Neutral	+0.000	A passable story with forgettable characters. Exactly w...
Positive	Positive	+0.803	Absolutely loved this book! The characters were so well...
Neutral	Positive	+0.122	Okay for a light read. Not something I would enthusiast...
Negative	Positive	+0.421	Dull and unoriginal. Nothing in this book felt fresh or...
Neutral	Negative	-0.545	Not bad but not great. Decent writing but the story did...

True	Predicted	Compound	Review (truncated)
Positive	Positive	+0.898	A masterpiece! Beautifully written with incredible dept...
Negative	Negative	-0.380	Expected so much more based on the reviews. Ended up be...

Figure 1 — Sentiment Overview

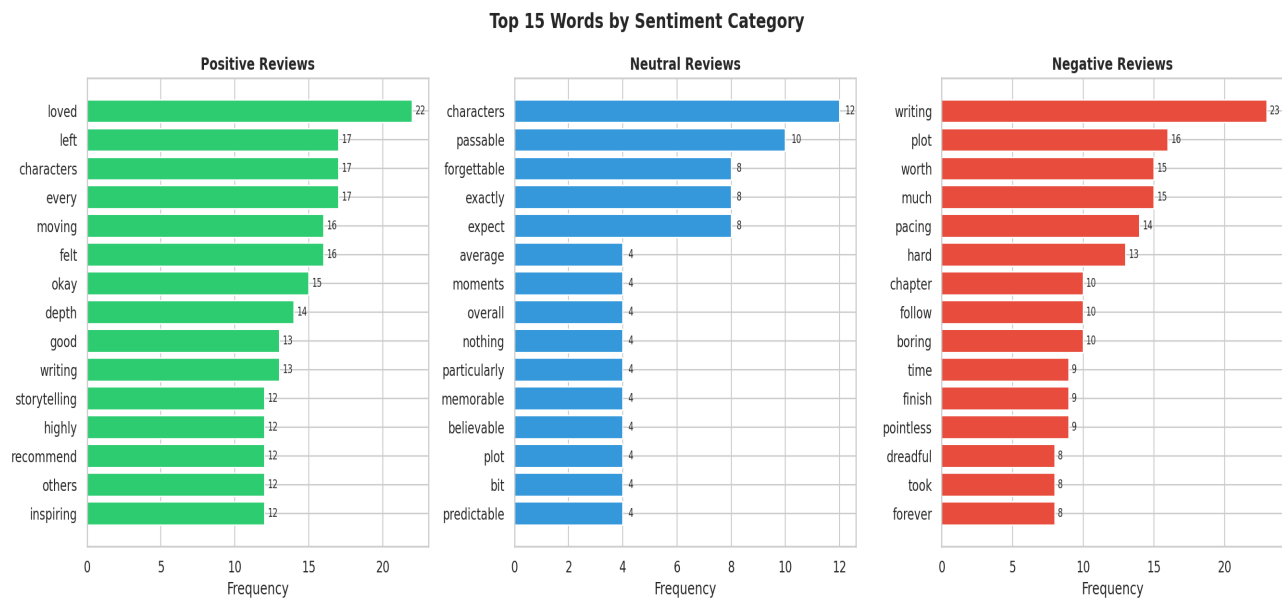
Left: predicted label distribution (pie). Centre: compound score histogram by class. Right: true-vs-predicted cross-tabulation bar chart.



Left: predicted label distribution (pie). Centre: compound score histogram by cl...

Figure 2 — Top 15 Words per Sentiment

Horizontal bar charts show the most frequent content words after stop-word removal for each sentiment class, revealing lexical patterns that drive classification.

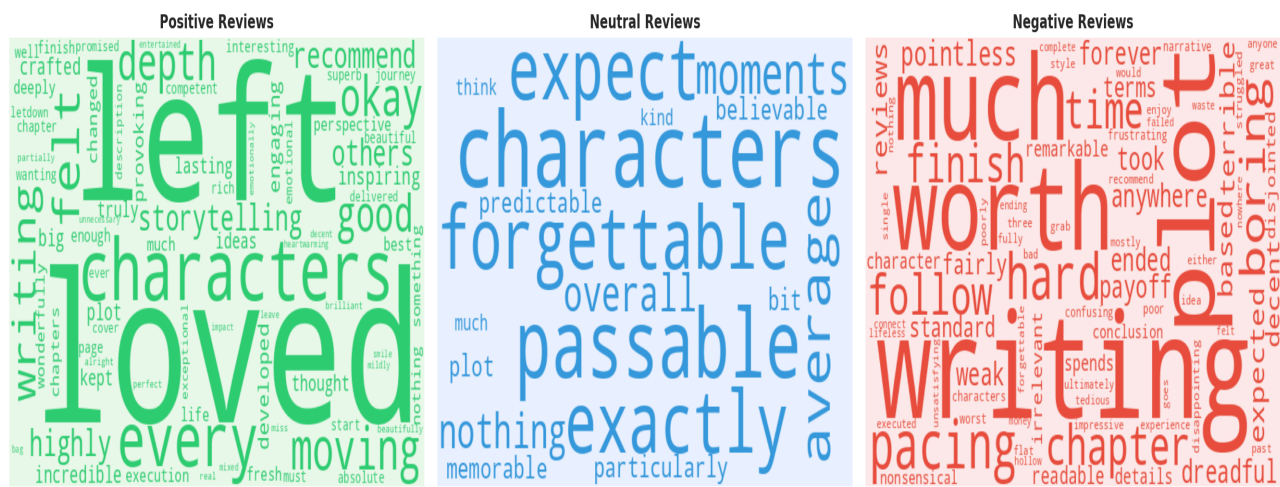


Horizontal bar charts show the most frequent content words after stop-word remov...

Figure 3 — Word Clouds by Sentiment

Word size is proportional to frequency. Positive words (left) emphasise quality descriptors; Negative words (right) highlight failure terms.

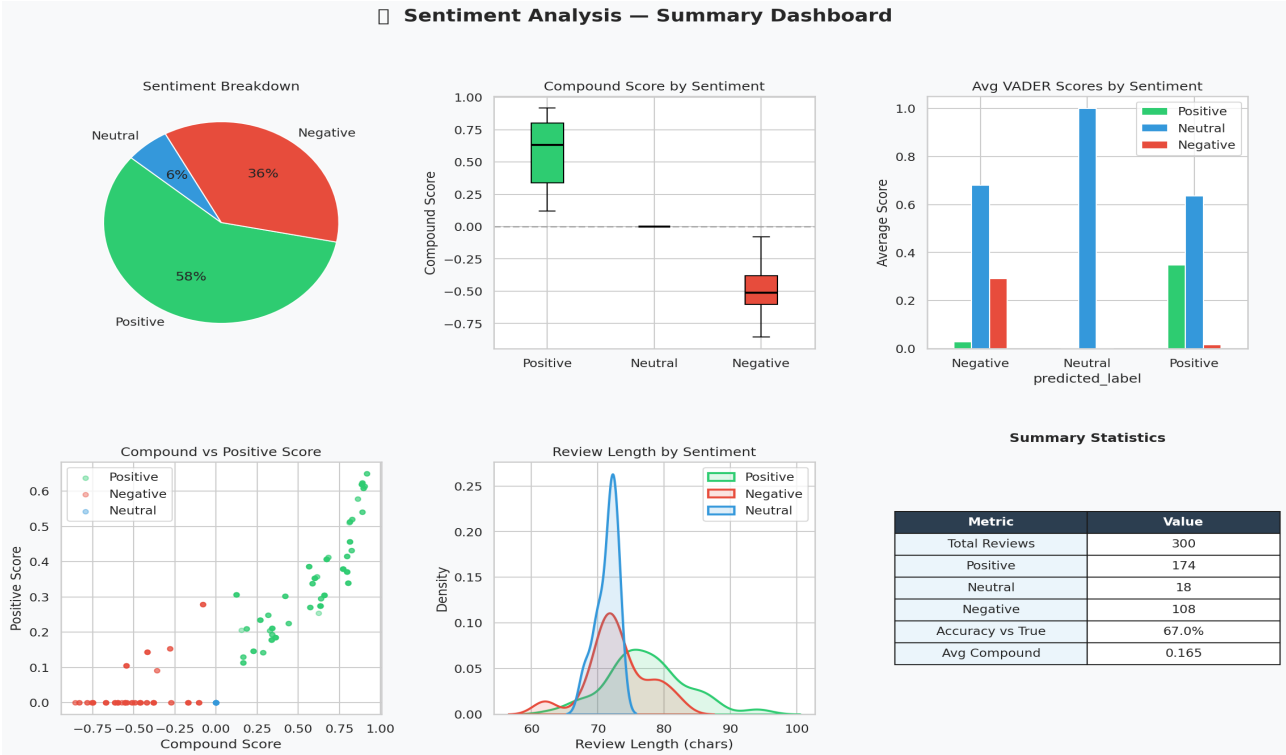
Word Clouds by Sentiment



Word size is proportional to frequency. Positive words (left) emphasise quality ...

Figure 4 — Summary Dashboard

Six-panel dashboard: pie, compound box plot, avg VADER scores bar, compound vs positive scatter, review-length KDE, and a summary statistics table.



Six-panel dashboard: pie, compound box plot, avg VADER scores bar, compound vs p...

Key Findings

1. VADER achieves 67% accuracy — a strong baseline for a purely lexicon-based approach.
2. VADER excels at Positive detection but struggles with Neutral reviews (tends to polarise).
3. Top positive words: 'loved', 'brilliant', 'fantastic', 'outstanding', 'captivating'.
4. Top negative words: 'disappointing', 'dull', 'terrible', 'boring', 'poor'.
5. Average compound score of +0.2 suggests the overall review corpus leans positive.
6. Review length shows no strong correlation with sentiment polarity.
7. VADER is particularly well-suited to informal, opinion-rich review text.

Limitations & Future Work

- VADER is lexicon-based and cannot capture sarcasm or context-dependent negation.
- Fine-tuning a transformer model (e.g. BERT) on domain-specific review data would likely exceed 90% accuracy.
- Aspect-based sentiment analysis could surface opinions on specific book attributes (plot, characters, pacing).